

AHMED FAWZY ABDELKADER

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Education

The American University in Cairo

B.Sc. Computer Engineering, Specialization in Artificial Intelligence; GPA: 3.8

Fall 2022 – Fall 2027

The Modern English School in Cairo

American High School Diploma; GPA: 4.2; National Honor Society, AP Scholar

Fall 2018 – Fall 2022

Work Experience

CIB

AI and Sustainable Finance Intern

Aug 2025

- Covered ESG risk, green banking, and AI-enabled financial use cases in a structured internship program.

Banque Misr 🌐

DevOps Intern

Aug 2024 – Oct 2024

- **Project – Cloud-Native DevOps Pipeline:** Provisioned AWS infrastructure with Terraform and deployed a Flask/SQLite app to EKS via Kubernetes manifests.
- Built a Jenkins pipeline with GitHub webhooks, SonarQube checks, and automated Docker/EKS rollouts.
- Configured Prometheus and Grafana dashboards for CPU, memory, disk, and network monitoring.

Skills

Languages: Python, JavaScript, Rust, C/C++, SQL, HTML/CSS, Bash

AI / ML: TensorFlow, PyTorch, scikit-learn, TinyML, pandas, NumPy

Web & Mobile: React, Django, Flask, Node.js, MongoDB, REST APIs

DevOps & Cloud: Docker, Kubernetes, AWS, Terraform, Jenkins, Prometheus, Grafana

Projects

TinyML-Based Intrusion Detection for Embedded IoT Security

Spring 2026 – Present

- Thesis: Building a TinyML anomaly-detection pipeline for resource-constrained IoT microcontrollers.
- Training and evaluating DSAEs, CNNs, RNNs, and classical ML models on large-scale IIoT network-flow datasets.
- Tech: Python, TensorFlow/Keras, PyTorch, scikit-learn, TinyML.

YOLO-Inspired Updates to RT-DETRv2 🌐

Spring 2026

- Applied YOLOv13-inspired HyperACE/FullPAD enhancements and knowledge distillation to RT-DETRv2.
- Benchmarked on MS COCO 2017 across AP metrics, FPS, latency, and parameter count.
- Tech: Python, PyTorch, CUDA.

Linux Process Manager 🌐

Fall 2025

- Built a Linux process tool with CLI and desktop GUI for inspection, control, and real-time CPU/memory monitoring.
- Tech: Rust, Tauri, React.

DigiMed 🌐

Fall 2024

- Full-stack medical platform for patient profiles, appointments, and medical records.
- Tech: Django, Flask, React, SQL.

Co-Curricular Activities

AUC Student Ambassadors Program

High Board Member, Work-Study Intern, and Student Ambassador

Oct 2023 – Present

- Managed ambassador operations across recruitment, onboarding, tour assignments, and event execution.
- Conducted ~50 campus tours for prospective students, international faculty, and VIP guests.
- **Project – Operations Tracking and Reporting System:** Built a Google Sheets workflow that reduced operational reporting from days to minutes.

Enactus AUC 🌐

Head of AI and Web Development Teams; Director of Projects

Fall 2024 – Fall 2026

- **Project – Text-to-3D Platform:** Led a five-member AI team to deliver a web application for image generation and 3D asset creation.
- Built MERN-stack APIs, Selenium pipelines, MongoDB Atlas storage, Docker services, and Azure VM deployments.